Mohammed Burhanuddeen R

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SENIOR QUALITY ENGINEER – 10 YEARS | MATERIALS/METALLURGY ENGINEER | QUALITY CONTROL/ASSURANCE | FERROUS/NON-FERROUS | HEAT TREATMENT | FORGINGS | NDT/DT/METALLURGICAL – TESTINGS | SUPPLIER QUALITY/EVALUATION | NEW PRODUCT DEVELOPMENT

CORE COMPETENCIES

- Heavy Open Die Forging
- Metallurgy
- Heat Treatment
- Destructive Testings
- Non Destructive Testings
- NPD, DFMA, PPAP
- Quality Management
- Best Practice Implementation
- Continual Improvement
- Exploratory Data Analysis
- Value Addition & Value Engineering (VA/VE)
- Supplier Quality Management
- Team Building & Leadership,
- Cross-functional Coordination
- Design of Experiments
- Hypothesis Testing
- Lean Management
- Statistical Modelling Tools

PROFILE SUMMARY

- **B.E. (Material Science / Metallurgical Engineering)** professional with **over 10 years of experience** spanning Heavy Open-Die Forging**,** NDT/DT**,** Supplier Development, New Product Development, Process Establishment, Quality Control, Quality Assurance & Data Analysis, Evaluating Vendor Performance and Metallurgical Testing, **with a key focus on Research & Development** and product development through business excellence philosophy.
- A critical thinker with an **adaptable mindset** and **highly analytical** data science experience to solve complex problems, passion for innovation and documented success in hands-on roles.
- Hands-on experience in **selection of Raw Material** for critical gear components and **Clean steel development using GRANTA.**
- **Proven expertise in** performing Strategic Steel Raw Material development and development of automotive steel grades.
- **Experienced in Forging process and their grades** in automobile and aerospace products.
- **Casting process established** right from the simulation to final quality of the product.
- **Heat Treatment cycle established** and hands-on experience in various Heat Treatment Processes and Conventional processes for various Steels, Stainless steel, Lightweight materials (especially Aluminum and Titanium based) & Superalloys (Nickel and Cobalt based).
- **Proficiency in designing & developing** new products in conformance to pre-defined technical specifications and requirements of the business/clients along with restructuring processes and improving operational efficiency
- **Resourceful in** modifying component design for manufacturability, scheduling jobs to the designers and ensuring on-time delivery**; ensuring the highest quality at the optimum production rate** and the most economical cost; controlling component rejections by conducting Failure Analysis
- **Specifying/recommending design modifications based on requirements.**
- **Feasibility study:** Dealing with various customer RFQ's (Request for Quotation) and releasing the technical review sheet for the same to marketing team with any queries (if have)
- Performing **Metallography tests** like general **microstructure**, grain size, depth of decarburization, inclusion content, alpha case (white layer), intergranular corrosion, carbide banding, intermetallic phases AND **Mechanical tests** like room temperature **Tensile**, **Microhardness**, **Rockwell** hardness, Compression, Hot Tensile, **Impact**, Fatigue, Fracture toughness, Stress Rupture.
- Creating **ITP**, **Work Instructions** and **Scan plans**; supervising overall Mechanical & NDE Level 2 technicians and planning activities for all **Destructive & Non-Destructive tests** and raise **NCR / CAPA / RCA** for non-conformance
- Creation and involvement of **PPAP**, **MPP** and **TMP** and releasing for production activities and Knowledge in **FMEA**, **TQM**, Six Sigma, **8D**, **PDCA**, **Quality Tools**, Lean Manufacturing and Quality systems.

WORK EXPERIENCE

Company - **AIMS GROUP :** ALIF – Jeddah, Kingdom of Saudi Arabia
Designation - Engineer - Quality Control and Assurance
Period - Since Mar 2025 to present

Company - **FOMAS GROUP :** Bay Forge Ltd - India & ASFO S.P.a – Italy **(Trained and worked in both locations)**
Designation - Senior Engineer - Quality Control and Assurance
Period - Feb 2017 to Mar 2025

Key Result Areas:

- Planning, organizing, directing, & controlling product development through business excellence philosophy.
- Evaluating vendor performance to ensure adherence to pre-defined specifications and supply of quality raw material/execution of job works w.r.t. service, cost, delivery & quality norms.
- Ensuring adherence to quality systems, maintaining requisite documents to ensure complete in-process quality & improvement in process capabilities.
- Reformulating existing products, ascertaining innovations in manufacturing process and setting up new projects
- Modifying component design for manufacturability, scheduling jobs to the designers and ensuring on-time delivery.

- Identifying opportunities through research on new processes and products using automation techniques
- Formulating technical requirements and specifying/recommending design modifications based on requirements; keeping a check on product quality & impact of design modifications on product features, machine & tooling /process
- Applying processing and assembly of product to ensure highest quality at the optimum production rate and the most economical cost; controlling component rejections.
- Prepared **ITP**, **Work Instructions** and **Scan plans,** and supervised overall Mechanical & NDT Level 2 technicians and planning activities for all **Destructive & Non-Destructive tests** and initiated **NCR / CAPA / RCA** for non-conformance
- Creation and involvement of **PPAP**, **MPP** and **TMP** and releasing for production activities and Knowledge in **FMEA**, **TQM**, Six Sigma, **8D**, **PDCA**, **Quality Tools**, Lean Manufacturing and Quality systems.

Highlights:
- Worked on **Automotive**, **Aerospace**, **O&G Material Grades** with renowned Indian and Italian steel mill like TATA, POSCO, Sunflag, Bhusan,Vardhman, JSW, Saarloha , Safau, Dal Zotto, Trinecke, Sotov, Lucchini and Asonext.
- Involved in **third-party inspection** and **Customer witnessing** activities for Forged components and liaised on Ferrous & Nonferrous Forging & Casting Part Development
- Conducted Failure Analysis of Crankshafts, Axle Beam, Connecting Rods, Steering Nuckles, Impellers (on Aluminum grade), Flaptrack (on Titanium grade)
- Data Science / Statistical Process Control / Supplier Quality project showcase [Click here]

ACHIEVEMENTS

PLM Implementation:
- Principal Architect during the implementation of Product Life Cycle Management software (SIEMENS - TeamCenter).
- Creating and maintaining all Quality Master Classes, Specification objects and Object structure and routing for the Quality cycle.
- Continual training of Quality assurance teams on working, releasing and creating certification booklets.

Cost Reduction:
- Overall, 15% reduction of Heat treatment expenses during 2021-2022 and 5% during 2023 by optimizing the heating rate and holding time through data analysis using statistical models.
- Reduction in quality claims from the customer by identifying the chokepoints and optimizing the overall certification activities.

Data Science:
- Exploring, analyzing and checking for deviations in incoming data regarding Raw material chemical, metallurgical and final product destructive testing datasets against company and international standards such as ASTM, SAE, ASM, ISO, DIN and EN norms, thereby giving insights on supplier and in-house quality standards.
- Customer data correlation from MDS creation to Quality reports and failure report analysis for error handling and highlight critical checkpoints and areas to improve using Regression, Clustering and Classification using K-means and DB Scans.
- Deployed a Recommendation engine that can search by material name to check a list of MDS or recommend a list of materials using testing specifications as a filtering guideline.

EDUCATION

Qualification	Institution	University / Board	Year	Percentage
B.E – Material Science \ Metallurgy	College of Engineering, Guindy Campus *(Top 20 in Engineering College, NIRF Ranking)*	Anna University	2012-2016	8 CGPA
Class 12	Velammal Matriculation Higher Secondary School	State Board	2010-2012	94.6%
Class 10	ICF Matriculation Higher Secondary School	State Board	2009-2010	92.6%

CERTIFICATIONS

- IBM Data Science Professional Certificate.
- Advanced Training on ASNT Level II NDT techniques
- ISO 9001:2015 Lead Auditor
- Heat Treat System Assessment - Special Process Auditor as per AIAG CQI 9 Version 4.
- Training on Six Sigma Black Belt.
- Master Diploma in Automotive Design.
- Training on 'AS9100:2016 (Rev D) & Internal Auditor Training'

SOFTWARES

- MsOffice, Citrix-BaaN, NetPro, SAP, Siemens PLM – Teamcenter
- AutoCAD, Creo, Catia, Granta, Ansys, NX-CAD, NX-CAE, NX-CAM, SolidWorks, KeyShot
- Minitab, Tableau, QlikView, Weka, Alteryx, IBM SPSS, Power BI
- Python modules : Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, Web Dashboarding : Streamlit

PERSONAL DETAILS

Date of Birth: 12th August 1994
Languages Known: English, Hindi, Urdu, Tamil & French
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